Fundamental dimensions

We have seen that given a $m \times n$ matrix, the column space, row space and null space are subspaces of $\mathbb{R}^n$, $\mathbb{R}^m$ and $\mathbb{R}^m$ respectively. In the previous examples, we have discussed how to find bases for these subspaces. Now, we define the dimension of these subspaces.

Definition of rank and nullity

Definition 1. Let A be an $m \times n$ matrix.

- The dimension of the column space of A is called the **rank** of A.
- The dimension of the null space of A is called the **nullity** of A.

Let us interpret these dimensions. Suppose the system $Ax = b$ is given.

- Rank tells us how many linearly independent columns (which can also be viewed as vectors) are there in A. Thus, rank gives us the exaxt number of columns (vectors) required to describe b, not one column (vector) less and not one column (vector) more.
- Nullity gives the number of free variables required to describe a solution of $Ax = \Theta$, (that is, when $b = \Theta$), where Θ is the zero vector of the underlying vector space.

We discuss how to find rank and nullity with the following two examples.

Example 1. Find rank and nullity of the matrix

$$A = \begin{pmatrix} 1 & 2 & 1 \\ -2 & -3 & 1 \\ 3 & 5 & 0 \end{pmatrix}$$

The method is to transform A into row reduced echelon form

$$\begin{pmatrix} 1 & 2 & 1 \\ -2 & -3 & 1 \\ 3 & 5 & 0 \end{pmatrix} \to \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 3 \\ 3 & 5 & 0 \end{pmatrix} \to \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 3 \\ 0 & -1 & -3 \end{pmatrix} \to \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{pmatrix}$$

- Rank of A is the number of non-zero rows in the row echelon form which is 2. This means that there are two linearly independent column vectors in A. This means a basis for the column space of A will contain only two vectors. *Using the previous examples for column space, can you identify those vectors?*
- For nullity, we use the row-reduced form to get the homogeneous system

  $$x + 2y + z = 0; \quad y + 3z = 0$$

  With $z = t \in \mathbb{R}$ as the free variable, the solution set is

  $$S = \{(5t, -3t, t) : t \in \mathbb{R}\} = \{t(5, -3, 1) : t \in \mathbb{R}\}$$

  Thus we require only one vector to describe any solution of $Ax = \Theta$ and hence nullity is 1. *Can you identify a basis for the null-space of A?*

Example 2. Let us find a basis and the dimension of the solution space of the homogeneous system

$$\begin{aligned} x_1 + 2x_2 + 2x_3 + 3x_4 &= 0 \\ 2x_1 + 4x_2 + x_3 + 3x_4 &= 0 \\ 3x_1 + 6x_2 + x_3 + 4x_4 &= 0 \end{aligned}$$

The first step is to identify the coefficient matrix A and write the system in the form $Ax = \Theta$. Since the given system is homogeneous, the solution space of the system of equation is the null-space of the coefficient matrix A.

We reduce the coefficient matrix to row echelon form

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 4 & 1 & 3 \\ 3 & 6 & 1 & 4 \end{pmatrix} \to \begin{pmatrix} 1 & 2 & 2 & 3 \\ 0 & 0 & -3 & -3 \\ 0 & 0 & -5 & -5 \end{pmatrix} \to \begin{pmatrix} 1 & 2 & 2 & 3 \\ 0 & 0 & -3 & -3 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

We thus obtain the equivalent system of equations

$$\begin{aligned} x_1 + 2x_2 + 2x_3 + 3x_4 &= 0 \\ -3x_3 - 3x_4 &= 0 \end{aligned}$$

Assuming $x_4 = t \in \mathbb{R}, x_2 = s \in \mathbb{R}$, the solution set is

$$S = \{(-2s - t, s, -t, t) : s, t \in \mathbb{R}\} = \{s(-2, 1, 0, 0) + t(-1, 0, -1, 1) : s, t \in \mathbb{R}\}.$$

Thus a basis for the null-space is $\mathcal{B} = \{(-2, 1, 0, 0), (-1, 0, -1, 1)\}$ and nullity is 2.

As an added information, the row-reduced form shows that the rank of the coefficient matrix A is also 2. Hence, even though there are 4 columns in A only two are required to describe any b such that $Ax = b$ has a solution. *Again, can you identify those two columns that will form a basis for the column space of A?*

0.1 The rank-nullity theorem

Consider the following matrix and its row echelon form

$$A = \begin{pmatrix} 2 & 5 & -3 & -4 & 8 \\ 4 & 7 & -4 & -3 & 9 \\ 6 & 9 & -5 & 2 & 4 \\ 0 & -9 & 6 & 5 & -6 \end{pmatrix} \sim \begin{pmatrix} 2 & 5 & -3 & -4 & 8 \\ 0 & -3 & 2 & 5 & -7 \\ 0 & 0 & 0 & 4 & -6 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Columns $1, 2, 4$ contain the pivot elements and hence the column space of A is of dimension 3 (which is also rank of A) with a basis $\{(2, 4, 6, 0), (5, 7, 9, -9), (-4, -3, 2, 5)\}$. Just like columns containing the pivot elements are related to rank and column space, the non-pivot columns are related to nullity and null-space. To see this, let us write the equations corresponding to the row echelon form

$$\begin{aligned} 2x_1 + 5x_2 - 3x_3 - 4x_4 + 8x_5 &= 0 \\ -3x_2 + 2x_3 + 5x_4 - 7x_5 &= 0 \\ 4x_4 - 6x_5 &= 0 \end{aligned}$$

To solve it, we let $x_5 = t$, (note that x_5 corresponds to a non-pivot column of A) and find x_4. From the second equation, we let $x_3 = s$ (again, x_3 corresponds to another non-pivot column of A) and find x_2. Finally, from the first equation we find x_1.

This is true for any matrix. That is, the non-pivot columns of a matrix will always correspond to the free variables in the null-space of the matrix. The number of non-pivot columns of a matrix will give the number of free variables in the solution set and hence the dimension of the null-space. Since the number of pivot columns (which gives the rank) plus the number of non-pivot columns (which gives the nullity) is exactly the number of columns of the given matrix, we have the following relation which is referred to as the rank-nullity theorem.

The rank-nullity theorem

If a matrix A has n columns then

$$\text{rank}(A) + \text{nullity}(A) = n$$